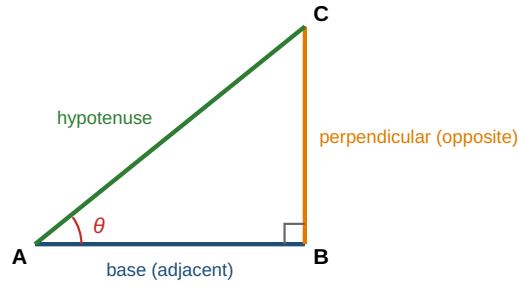


Introduction to Trigonometry

Trigonometric ratios, standard angles, complementary angles and identities



What is inside

1. Why trigonometry? Where does it come from?
2. Naming the sides of a right triangle
3. The six trigonometric ratios
4. Ratios depend only on the angle, not the size
5. Finding all ratios when one is known
6. Standard angles: 0° , 30° , 45° , 60° , 90°
7. Complementary angles
8. Trigonometric identities
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10. Exercises (Levels A–E) with answers
11. One-page formula sheet

Himanshu Malik
PGT Mathematics, Rohtak

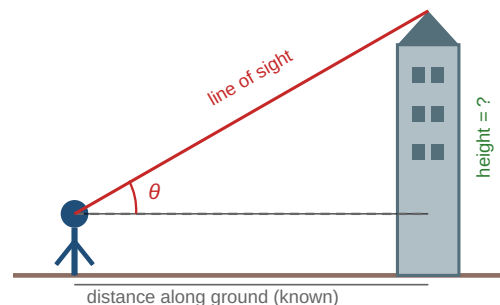
1. Why trigonometry?

Look at the picture. A student stands some distance from a tower and looks up at its top. Two things are easy to measure: the **distance along the ground** and the **angle** at which the student looks up (with a simple instrument called a clinometer, or even a protractor).

One thing is **not** easy to measure: the **height** of the tower.

Trigonometry is the bridge between them. If we know one side and one angle of a right triangle, trigonometry lets us calculate the other sides. The word itself is Greek: *tri* (three) + *gon* (sides) + *metron* (measure), so it literally means "measuring triangles".

This idea was used by ancient astronomers to find distances to the Moon and planets, by the Indian mathematician **Aryabhata** (5th century CE) who first tabulated values of the sine (he called it *jya*), and today by engineers, architects, pilots, GPS satellites and video-game designers.

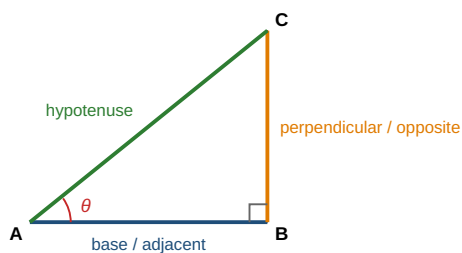


Angle θ and the ground distance are known; the height is what we want.

In this chapter we study only **right-angled triangles** and **acute angles** (between 0° and 90°). Everything we learn is built from one triangle and one angle in it.

2. Naming the sides of a right triangle

Take a triangle ABC, right-angled at B. Pick one of the acute angles, say the angle at A, and call it θ (theta). The three sides now get **names relative to θ** :



Sides named with respect to angle A

Hypotenuse (H) – the side opposite the right angle. Always the longest side. It never changes, whichever acute angle you pick.

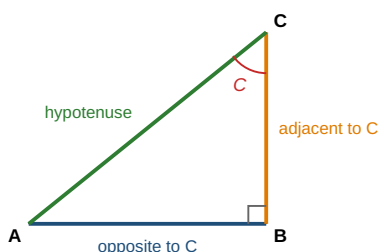
Perpendicular / Opposite (P) – the side *facing* the angle θ . It does not touch θ .

Base / Adjacent (B) – the side that *touches* θ but is not the hypotenuse. "Adjacent" means "next to".

NCERT uses the words *opposite* and *adjacent*; many books (and the popular Hindi mnemonic) use *perpendicular* and *base*. Both mean the same thing.

The names change when the angle changes

This is the single most common source of mistakes. The **same** triangle, viewed from angle C instead of angle A, swaps the roles of the two shorter sides:



Sides named with respect to angle C

Side	For angle A	For angle C
AC	hypotenuse	hypotenuse
BC	opposite (P)	adjacent (B)
AB	adjacent (B)	opposite (P)

Rule of thumb: first fix the angle, *then* name the sides. Never name the sides first.

3. The six trigonometric ratios

A **trigonometric ratio** is simply one side of the right triangle divided by another. With three sides there are six possible ratios, and each has a name. For the angle θ at A in the triangle above:

Ratio	Full name	Definition	In words
$\sin \theta$	sine	$\frac{P}{H} = \frac{BC}{AC}$	opposite $\div$ hypotenuse
$\cos \theta$	cosine	$\frac{B}{H} = \frac{AB}{AC}$	adjacent $\div$ hypotenuse
$\tan \theta$	tangent	$\frac{P}{B} = \frac{BC}{AB}$	opposite $\div$ adjacent
$\operatorname{cosec} \theta$	cosecant	$\frac{H}{P} = \frac{1}{\sin \theta}$	reciprocal of sin
$\sec \theta$	secant	$\frac{H}{B} = \frac{1}{\cos \theta}$	reciprocal of cos
$\cot \theta$	cotangent	$\frac{B}{P} = \frac{1}{\tan \theta}$	reciprocal of tan

Memory trick 1 (English): SOH – CAH – TOA

Sin = Opposite / Hypotenuse

Cos = Adjacent / Hypotenuse

Tan = Opposite / Adjacent

Memory trick 2 (Hindi): "Pandit Badri Prasad, Har Har Bole"

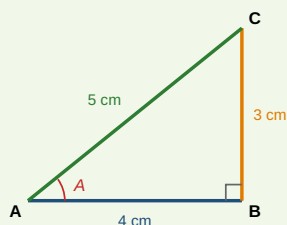
sin = P/H, cos = B/H, tan = P/B

(Pandit–Har, Badri–Har, Prasad–Bole). The other three are just the reciprocals.

Two relations you will use constantly: $\tan \theta = \frac{\sin \theta}{\cos \theta}$ $\cot \theta = \frac{\cos \theta}{\sin \theta}$

Check: $\sin \theta \div \cos \theta = (P/H) \div (B/H) = P/B = \tan \theta$.

Example 1



In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 4$ cm, $BC = 3$ cm. Find all six ratios of angle A.

Step 1 – find the third side. By Pythagoras, $AC^2 = AB^2 + BC^2 = 16 + 9 = 25$, so $AC = 5$ cm.

Step 2 – name the sides for angle A. H = AC = 5, P = BC = 3, B = AB = 4.

Step 3 – write the ratios.

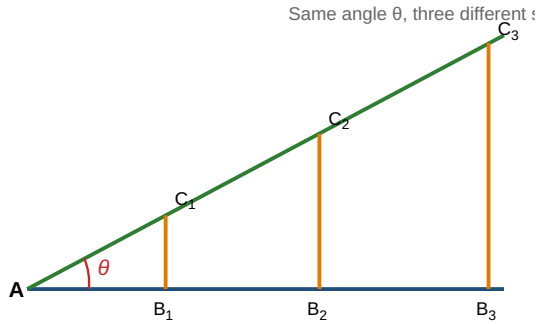
$$\sin A = \frac{3}{5}, \quad \cos A = \frac{4}{5}, \quad \tan A = \frac{3}{4}, \quad \operatorname{cosec} A = \frac{5}{3}, \quad \sec A = \frac{5}{4}, \quad \cot A = \frac{4}{3}$$

Try it: now do the same for angle C. You should get $\sin C = 4/5$ and $\cos C = 3/5$ – the sine and cosine swap!

Watch out

"sin θ " is one symbol, not "sin $\times \theta$ ". You cannot cancel the "sin". Also, a trigonometric ratio has **no units** – it is a length divided by a length, so the cm cancel out.

4. The ratios depend only on the angle, not on the size



Draw an angle θ at A. From different points B_1, B_2, B_3 on one arm, drop perpendiculars to meet the other arm at C_1, C_2, C_3 .

The three triangles $AB_1C_1, AB_2C_2, AB_3C_3$ all have the same angles ($\theta, 90^\circ$, and $90^\circ - \theta$), so they are **similar** (AA similarity, Chapter 6). In similar triangles corresponding sides are in the same ratio, so

$$\frac{B_1C_1}{AC_1} = \frac{B_2C_2}{AC_2} = \frac{B_3C_3}{AC_3} = \sin \theta$$

Conclusion: $\sin \theta, \cos \theta, \tan \theta \dots$ are **numbers that belong to the angle θ** . Any right triangle containing θ will give the same value, no matter how big or small it is. That is exactly why we can make a table of values (Section 6) and use it for every triangle in the world.

How big can a ratio be?

The hypotenuse is always the longest side, so $P < H$ and $B < H$. Therefore

$$0 < \sin \theta < 1 \quad 0 < \cos \theta < 1 \quad \operatorname{cosec} \theta > 1 \quad \sec \theta > 1 \quad (\text{for } 0^\circ < \theta < 90^\circ)$$

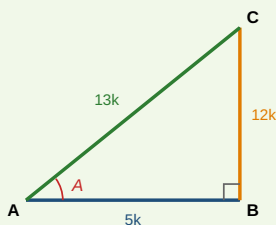
$\tan \theta$ and $\cot \theta$ can be any positive number – $\tan \theta$ becomes very large as θ approaches 90° .

So if a question asks "is $\sin A = 5/3$ possible?", the answer is no, without any calculation.

5. Finding all the ratios when one is known

If one ratio is given, draw a right triangle, label two sides using that ratio, find the third side by Pythagoras, then read off everything else.

Example 2



If $\cos A = \frac{5}{13}$, find $\sin A$ and $\tan A$.

$\cos A = B/H = 5/13$. This does not mean B is exactly 5 – only that the ratio is 5 : 13. So take **$AB = 5k$** and **$AC = 13k$** for some positive number k .

By Pythagoras: $BC^2 = (13k)^2 - (5k)^2 = 169k^2 - 25k^2 = 144k^2$, so **$BC = 12k$** .

$$\sin A = \frac{12k}{13k} = \frac{12}{13}, \quad \tan A = \frac{12k}{5k} = \frac{12}{5}$$

The k always cancels – that is Section 4 in action. In an exam you may write "let $AB = 5k$ " or simply "let $AB = 5, AC = 13$ "; both are accepted, but the k -method shows you understand why.

Example 3

If $15 \cot A = 8$, find $\sin A$ and $\sec A$.

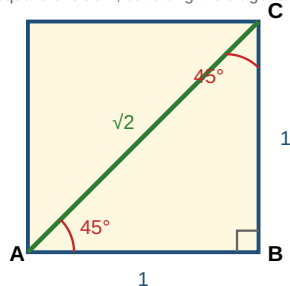
$\cot A = 8/15 = B/P$, so let $B = 8k, P = 15k$. Then $H^2 = 64k^2 + 225k^2 = 289k^2, H = 17k$.

$$\sin A = P/H = \frac{15}{17}, \quad \sec A = H/B = \frac{17}{8}$$

6. Standard angles: 0°, 30°, 45°, 60°, 90°

For five special angles the ratios are exact, simple numbers. They are not magic – they come from two triangles you can draw yourself.

square of side 1, cut along the diagonal

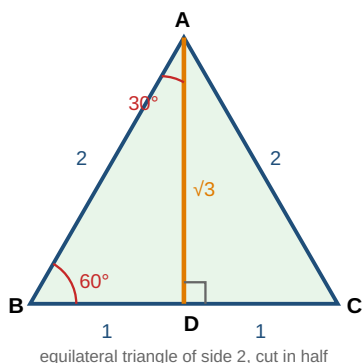


45°: half a square

Cut a square of side 1 along its diagonal. Each half is a right triangle with two 45° angles and legs 1 and 1. By Pythagoras the hypotenuse is $\sqrt{1 + 1} = \sqrt{2}$.

$$\sin 45^\circ = \frac{1}{\sqrt{2}}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}}, \quad \tan 45^\circ = \frac{1}{1} = 1$$

$\sin 45^\circ = \cos 45^\circ$ because the two legs are equal – the triangle looks the same from both acute angles.



30° and 60°: half an equilateral triangle

Take an equilateral triangle ABC of side 2 (all angles 60°). Drop the perpendicular AD from A. It bisects BC (so BD = 1) and bisects ∠A (so ∠BAD = 30°). By Pythagoras, $AD = \sqrt{4 - 1} = \sqrt{3}$.

In $\triangle ABD$: hypotenuse AB = 2, the side opposite 30° is BD = 1, the side opposite 60° is AD = $\sqrt{3}$.

$$\sin 30^\circ = \frac{1}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}, \quad \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \cos 60^\circ = \frac{1}{2}, \quad \tan 60^\circ = \sqrt{3}$$

0° and 90°: what happens at the ends

Imagine θ shrinking towards 0°. The perpendicular P shrinks to nothing while the base B becomes equal to H. So **$\sin 0^\circ = 0$, $\cos 0^\circ = 1$, $\tan 0^\circ = 0$** . Now let θ grow towards 90°: P becomes equal to H and B shrinks to nothing. So **$\sin 90^\circ = 1$, $\cos 90^\circ = 0$** , and $\tan 90^\circ = P/B$ would mean dividing by zero – it is **not defined**.

The table (learn it by heart)

θ	0°	30°	45°	60°	90°
sin θ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan θ	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined
cosec θ	not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec θ	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	not defined
cot θ	not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Memory trick for the table. Write 0, 1, 2, 3, 4. Take the square root and divide by 2:

$$\sin: \frac{\sqrt{0}}{2} = 0, \quad \frac{\sqrt{1}}{2} = \frac{1}{2}, \quad \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}, \quad \frac{\sqrt{3}}{2}, \quad \frac{\sqrt{4}}{2} = 1$$

cos is the same row written backwards. **tan** = sin ÷ cos. The last three rows are reciprocals of the first three. So you only ever memorise "0 1 2 3 4".

Example 4

Evaluate: (i) $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ (ii) $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$

$$(i) = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1 \quad (\text{Notice: this equals } \sin 90^\circ. \text{ Not a coincidence – you will meet } \sin(A + B) \text{ in Class 11.})$$

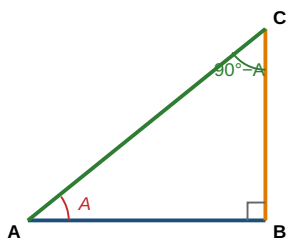
$$(ii) = 2(1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = 2 + \frac{3}{4} - \frac{3}{4} = 2$$

Example 5 – finding the angle

If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B.

From the table, $\tan 60^\circ = \sqrt{3}$, so $A + B = 60^\circ$. Also $\tan 30^\circ = 1/\sqrt{3}$, so $A - B = 30^\circ$.

Adding: $2A = 90^\circ$, $A = 45^\circ$. Subtracting: $2B = 30^\circ$, $B = 15^\circ$.

7. Complementary angles

$$\angle A + \angle C = 90^\circ, \text{ so } \angle C = 90^\circ - A$$

Two angles are **complementary** if they add up to 90° . In a right triangle the two acute angles are always complementary: since $\angle B = 90^\circ$, $\angle A + \angle C = 90^\circ$, i.e. $\angle C = 90^\circ - A$.

Now look at the sides. For angle A: opposite = BC, adjacent = AB. For angle C: opposite = AB, adjacent = BC. The roles are swapped. Therefore

$$\sin(90^\circ - A) = \sin C = \frac{AB}{AC} = \cos A$$

and similarly for the others. This gives us six results:

$\sin(90^\circ - A) = \cos A$	$\cos(90^\circ - A) = \sin A$	$\tan(90^\circ - A) = \cot A$
$\operatorname{cosec}(90^\circ - A) = \sec A$	$\sec(90^\circ - A) = \operatorname{cosec} A$	$\cot(90^\circ - A) = \tan A$

How to remember: the "co" in **cosine**, **cosecant** and **cotangent** stands for **complementary**. Each "co" function of an angle equals the plain function of its complement. Check with the table: $\sin 30^\circ = 1/2 = \cos 60^\circ$.

Example 6

Evaluate: (i) $\frac{\sin 18^\circ}{\cos 72^\circ}$ (ii) $\cos 48^\circ - \sin 42^\circ$ (iii) $\tan 26^\circ \cdot \tan 64^\circ$

(i) $72^\circ = 90^\circ - 18^\circ$, so $\cos 72^\circ = \cos(90^\circ - 18^\circ) = \sin 18^\circ$. The fraction is $\sin 18^\circ / \sin 18^\circ = 1$.

(ii) $\cos 48^\circ = \cos(90^\circ - 42^\circ) = \sin 42^\circ$. So $\cos 48^\circ - \sin 42^\circ = 0$. Answer: **0**.

(iii) $\tan 64^\circ = \tan(90^\circ - 26^\circ) = \cot 26^\circ = 1/\tan 26^\circ$. Product = $\tan 26^\circ \times (1/\tan 26^\circ) = 1$.

Example 7

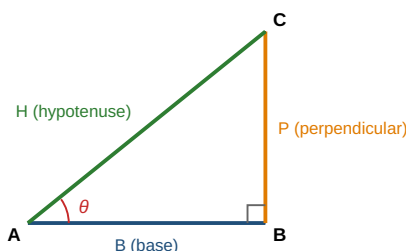
If $\tan 2A = \cot(A - 18^\circ)$, where $2A$ is an acute angle, find A.

$\cot(A - 18^\circ) = \tan(90^\circ - (A - 18^\circ)) = \tan(108^\circ - A)$. So $\tan 2A = \tan(108^\circ - A)$, giving $2A = 108^\circ - A$, $3A = 108^\circ$, $A = 36^\circ$.

Shortcut: when $\tan X = \cot Y$ (or $\sin X = \cos Y$, or $\sec X = \operatorname{cosec} Y$), you can simply write $X + Y = 90^\circ$. Here $2A + A - 18^\circ = 90^\circ$.

8. Trigonometric identities

An **identity** is an equation that is true for **every** value of the angle (unlike an ordinary equation like $\tan \theta = 1$, which is true only for $\theta = 45^\circ$). There are three fundamental identities, and all three come from Pythagoras.



In the triangle, Pythagoras says $P^2 + B^2 = H^2$.

Divide every term by H^2 :

$$\frac{P^2}{H^2} + \frac{B^2}{H^2} = 1 \implies \sin^2\theta + \cos^2\theta = 1 \quad \dots \text{(I)}$$

Instead divide by B^2 :

$$\frac{P^2}{B^2} + 1 = \frac{H^2}{B^2} \implies 1 + \tan^2\theta = \sec^2\theta \quad \dots \text{(II)}$$

Or divide by P^2 :

$$1 + \frac{B^2}{P^2} = \frac{H^2}{P^2} \implies 1 + \cot^2\theta = \operatorname{cosec}^2\theta \quad \dots \text{(III)}$$

Notation: $\sin^2\theta$ means $(\sin \theta)^2$, i.e. $(\sin \theta) \times (\sin \theta)$. It does **not** mean $\sin(\theta^2)$ and it does not mean $\sin(\sin \theta)$.

Useful rearrangements: $\sin^2\theta = 1 - \cos^2\theta = (1 - \cos \theta)(1 + \cos \theta)$, $\sec^2\theta - \tan^2\theta = 1$, $\operatorname{cosec}^2\theta - \cot^2\theta = 1$.

How to prove an identity – the method

1. Start from the **more complicated** side (usually the LHS). Never move terms across the "=" sign – you are not solving an equation.
2. Convert everything to **sin and cos** if you are stuck. ($\tan = \sin/\cos$, $\sec = 1/\cos$, ...)
3. Use the three identities and ordinary algebra: take LCM, factorise, cancel.
4. Keep going until you reach exactly the other side. Write "= RHS" at the end.

Example 8

Prove that $(1 + \tan^2 A)(1 - \sin A)(1 + \sin A) = 1$.

$$\begin{aligned} \text{LHS} &= \sec^2 A \cdot (1 - \sin A)(1 + \sin A) \quad [\text{by identity II}] \\ &= \sec^2 A \cdot (1 - \sin^2 A) \quad [(a - b)(a + b) = a^2 - b^2] \\ &= \sec^2 A \cdot \cos^2 A \quad [\text{by identity I}] \\ &= \frac{1}{\cos^2 A} \cdot \cos^2 A = 1 = \text{RHS.} \end{aligned}$$

Example 9

Prove that $(\operatorname{cosec} A - \cot A)^2 = \frac{1 - \cos A}{1 + \cos A}$.

$$\text{LHS} = \left(\frac{1}{\sin A} - \frac{\cos A}{\sin A} \right)^2 = \left(\frac{1 - \cos A}{\sin A} \right)^2 = \frac{(1 - \cos A)^2}{\sin^2 A} = \frac{(1 - \cos A)^2}{1 - \cos^2 A} = \frac{(1 - \cos A)(1 - \cos A)}{(1 - \cos A)(1 + \cos A)} = \frac{1 - \cos A}{1 + \cos A} = \text{RHS.}$$

Example 10 – using an identity to find a value

If $\sin \theta + \cos \theta = \sqrt{2} \cos \theta$ ($\theta \neq 90^\circ$), show that $\cot \theta = \sqrt{2} + 1$.

$\sin \theta = \sqrt{2} \cos \theta - \cos \theta = (\sqrt{2} - 1) \cos \theta$. Dividing by $\cos \theta$: $\tan \theta = \sqrt{2} - 1$.

$$\cot \theta = \frac{1}{\sqrt{2} - 1} = \frac{1}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1} = \frac{\sqrt{2} + 1}{2 - 1} = \sqrt{2} + 1.$$

9. Common mistakes to avoid

Mistake 1 Naming the sides before choosing the angle. "Opposite" and "adjacent" are meaningless until you say *opposite to which angle*.

Mistake 2 Writing $\sin A = 5/3$ or $\sec A = 12/13$. $\sin$ and $\cos$ are never more than 1; $\sec$ and cosec are never less than 1.

Mistake 3 Cancelling "sin" as if it were a number: $\frac{\sin A}{\sin B} \neq \frac{A}{B}$. Likewise $\sin(A + B) \neq \sin A + \sin B$. Check with the table: $\sin 60^\circ \neq \sin 30^\circ + \sin 30^\circ$.

Mistake 4 Confusing $\sin^2 A$ with $\sin A^2$. In this chapter you will only ever see $\sin^2 A = (\sin A)^2$.

Mistake 5 Using $\tan 90^\circ$ or $\sec 90^\circ$ in a calculation. They are not defined. If a question seems to need them, re-read the question.

Mistake 6 Leaving $\sqrt{2}$ or $\sqrt{3}$ in the denominator when a "simplified" answer is asked. Rationalise: $1/\sqrt{2} = \sqrt{2}/2$.

Mistake 7 In identity proofs, moving terms from one side to the other. Work on one side only until it becomes the other.

10. Exercises

Work in your notebook. Draw a triangle for every question in Levels A and E. Answers are on the next page – check only after you have tried.

Level A – Ratios from a triangle

1. In $\triangle PQR$, right-angled at Q , $PQ = 12$ cm and $QR = 5$ cm. Find $\sin P$, $\cos P$, $\tan P$, $\sin R$ and $\cos R$.
2. If $\sin A = \frac{3}{4}$, find $\cos A$ and $\tan A$.
3. If $\cot \theta = \frac{7}{8}$, evaluate $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$.
4. State whether true or false, with a reason:
 - a. $\sin \theta = \frac{4}{3}$ for some acute angle θ .
 - b. $\cos A$ is the abbreviation for $\cos \times A$.
 - c. $\tan A = \frac{12}{5}$ for some acute angle A .
 - d. $\sec A = \frac{12}{13}$ for some acute angle A .
 - e. $\sin \theta = \cos \theta$ when $\theta = 45^\circ$.

Level B – Standard angles

5. Evaluate:
 - a. $\sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$
 - b. $4 \cot^2 45^\circ - \sec^2 60^\circ + \sin^2 60^\circ + \cos^2 90^\circ$
 - c. $\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$
 - d. $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$
6. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, with $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .
7. If $2 \sin 2\theta = \sqrt{3}$ and 2θ is acute, find θ .
8. Is it true that $\sin 60^\circ = 2 \sin 30^\circ$? Verify using the table.

Level C – Complementary angles

9. Evaluate: (a) $\frac{\sin 25^\circ}{\cos 65^\circ}$ (b) $\tan 48^\circ \tan 23^\circ \tan 42^\circ \tan 67^\circ$ (c) $\cos 38^\circ \cos 52^\circ - \sin 38^\circ \sin 52^\circ$
10. If $\sec 4A = \operatorname{cosec}(A - 20^\circ)$, where $4A$ is an acute angle, find A .
11. Express $\sin 67^\circ + \cos 75^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .
12. If A , B and C are the angles of a triangle, show that $\sin \frac{B+C}{2} = \cos \frac{A}{2}$.

Level D – Identities

13. Prove the following:

a. $(\sec A + \tan A)(1 - \sin A) = \cos A$

b. $\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$

c. $\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$

d. $\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A$

e. $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \frac{1}{\tan A + \cot A}$

14. If $x = a \cos \theta + b \sin \theta$ and $y = a \sin \theta - b \cos \theta$, prove that $x^2 + y^2 = a^2 + b^2$.

15. If $\sin \theta = \frac{12}{13}$, find the value of $\frac{\sin^2 \theta - \cos^2 \theta}{2 \sin \theta \cos \theta} \times \frac{1}{\tan^2 \theta}$.

Level E – Board-style / application

16. In $\triangle ABC$, right-angled at C, $\tan A = \frac{1}{\sqrt{3}}$. Find the value of $\sin A \cos B + \cos A \sin B$.

17. A 10 m ladder leans against a wall and makes an angle of 60° with the ground. How high up the wall does the ladder reach? (Use $\sqrt{3} \approx 1.73$.)

18. In $\triangle ABC$ right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Show that $\sin^2 A + \cos^2 A = 1$ by actually calculating both ratios.

19. If $\sqrt{3} \tan \theta = 3 \sin \theta$ ($\theta \neq 0^\circ$), find $\sin^2 \theta - \cos^2 \theta$.

20. **Challenge.** Without a calculator, decide which is larger: $\sin 40^\circ$ or $\cos 40^\circ$. Explain using the triangle, not the table.

Answers and hints

Level A

1. $PR = 13$ cm. $\sin P = 5/13$, $\cos P = 12/13$, $\tan P = 5/12$; $\sin R = 12/13$, $\cos R = 5/13$.
2. $P = 3k$, $H = 4k$, $B = \sqrt{7}k$. $\cos A = \sqrt{7}/4$, $\tan A = 3/\sqrt{7} = 3\sqrt{7}/7$.
3. The expression $= (1 - \sin^2\theta)/(1 - \cos^2\theta) = \cos^2\theta/\sin^2\theta = \cot^2\theta = 49/64$.
4. (a) False – $\sin \theta \leq 1$. (b) False – $\cos A$ is a single symbol. (c) True – $\tan$ can be any positive number. (d) False – $\sec A \geq 1$. (e) True – both equal $1/\sqrt{2}$.

Level B

5. (a) $(1 + \sqrt{3})/(2\sqrt{2}) = (\sqrt{2} + \sqrt{6})/4$. (b) $4 - 4 + 3/4 + 0 = 3/4$. (c) $(3\sqrt{2} - \sqrt{6})/8$. (d) $(5/4 + 16/3 - 1)/1 = 67/12$.
6. $A - B = 30^\circ$, $A + B = 60^\circ$; $A = 45^\circ$, $B = 15^\circ$.
7. $\sin 2\theta = \sqrt{3}/2$, so $2\theta = 60^\circ$, $\theta = 30^\circ$.
8. False. $\sin 60^\circ = \sqrt{3}/2 \approx 0.87$ but $2 \sin 30^\circ = 1$. (This is Mistake 3 again: $\sin(2 \times 30^\circ) \neq 2 \sin 30^\circ$.)

Level C

9. (a) 1. (b) $\tan 48^\circ \tan 42^\circ = 1$ and $\tan 23^\circ \tan 67^\circ = 1$, so 1. (c) $\cos 52^\circ = \sin 38^\circ$ and $\sin 52^\circ = \cos 38^\circ$; the expression is $\cos 38^\circ \sin 38^\circ - \sin 38^\circ \cos 38^\circ = 0$.
10. $4A + (A - 20^\circ) = 90^\circ$, so $A = 22^\circ$.
11. $\cos 23^\circ + \sin 15^\circ$.
12. $A + B + C = 180^\circ$, so $(B + C)/2 = 90^\circ - A/2$. Then $\sin(90^\circ - A/2) = \cos(A/2)$.

Level D

13. (a) $LHS = (1 + \sin A)(1 - \sin A)/\cos A = \cos^2 A/\cos A = \cos A$. (b) Take LCM: $(\sin^2 A + (1 + \cos A)^2)/(\sin A (1 + \cos A)) = (2 + 2 \cos A)/(\sin A (1 + \cos A)) = 2/\sin A$. (c) Factor: $\sin \theta (1 - 2 \sin^2 \theta)/(\cos \theta (2 \cos^2 \theta - 1))$; note $1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$. (d) Multiply inside the root by $(1 + \sin A)/(1 + \sin A)$: $(1 + \sin A)^2/\cos^2 A$, take the root. (e) $LHS = (\cos^2 A/\sin A)(\sin^2 A/\cos A) = \sin A \cos A$; $RHS = 1/((\sin^2 A + \cos^2 A)/(\sin A \cos A)) = \sin A \cos A$.
14. Expand both squares; the cross terms $2ab \sin \theta \cos \theta$ cancel, and $(a^2 + b^2)(\cos^2 \theta + \sin^2 \theta) = a^2 + b^2$.
15. $\cos \theta = 5/13$, $\tan \theta = 12/5$. Value $= (119/169)/(120/169) \times 25/144 = 119/120 \times 25/144 = 595/3456$.

Level E

16. $\tan A = 1/\sqrt{3}$ gives $A = 30^\circ$, so $B = 60^\circ$. Value $= (1/2)(1/2) + (\sqrt{3}/2)(\sqrt{3}/2) = 1$. (Or: $B = 90^\circ - A$, so the expression is $\sin^2 A + \cos^2 A = 1$.)
17. Height $= 10 \sin 60^\circ = 10 \times \sqrt{3}/2 = 5\sqrt{3} \approx 8.65$ m.
18. $AC = 25$ cm. $\sin A = 7/25$, $\cos A = 24/25$; $49/625 + 576/625 = 625/625 = 1$.
19. $\sqrt{3} \sin \theta/\cos \theta = 3 \sin \theta$, so $\cos \theta = \sqrt{3}/3 = 1/\sqrt{3}$, $\cos^2 \theta = 1/3$, $\sin^2 \theta = 2/3$. Answer: $1/3$.
20. $\cos 40^\circ = \sin 50^\circ$. In a right triangle the larger angle faces the longer side, so the side opposite 50° is longer than the side opposite 40° ; hence $\sin 50^\circ > \sin 40^\circ$, i.e. **$\cos 40^\circ$ is larger**. (In general $\sin \theta$ increases and $\cos \theta$ decreases as θ goes from 0° to 90° .)

11. One-page formula sheet

Definitions (angle θ , right angle elsewhere)

$$\sin \theta = \frac{P}{H} \quad \cos \theta = \frac{B}{H} \quad \tan \theta = \frac{P}{B}$$

$$\operatorname{cosec} \theta = \frac{H}{P} \quad \sec \theta = \frac{H}{B} \quad \cot \theta = \frac{B}{P}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Range ($0^\circ < \theta < 90^\circ$)

$$0 < \sin \theta, \cos \theta < 1 \quad \sec \theta, \operatorname{cosec} \theta > 1 \quad \tan \theta, \cot \theta > 0$$

Complementary angles

$$\sin(90^\circ - A) = \cos A \quad \cos(90^\circ - A) = \sin A$$

$$\tan(90^\circ - A) = \cot A \quad \cot(90^\circ - A) = \tan A$$

$$\sec(90^\circ - A) = \operatorname{cosec} A \quad \operatorname{cosec}(90^\circ - A) = \sec A$$

Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—
cosec	—	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	—
cot	—	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

— means "not defined"

Remember: sin row = $\sqrt{0}, \sqrt{1}, \sqrt{2}, \sqrt{3}, \sqrt{4}$ all over 2. cos row = sin row reversed. tan = sin $\div$ cos.

Proving identities: start from the complicated side · convert to sin and cos · LCM, factorise, cancel · never cross the "=" sign.

